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UNIT-5

BASICS OF PHYTOCHEMISTRY

- i) Extraction - all techniques.
- ii) Applications of latest technique like spectroscopy, chromatography, Electrophoresis in isolation, purification and identification of crude drugs.

ImpQus →

- ① Isolation, purification and identification of crude drugs by the application of latest technique

~~chromatography~~

~~spectroscopy~~

- ② Write a note on extraction & its method!—

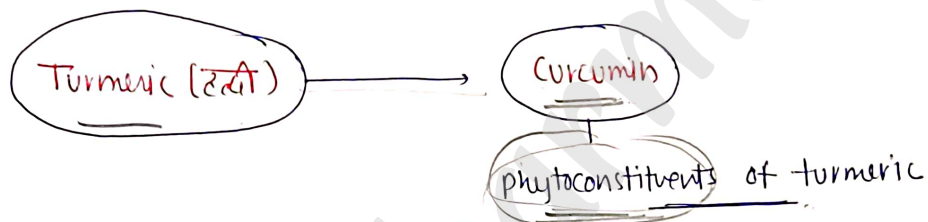
- percolation - soxhlet process - PLC

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• Phytoconstituents →

These are chemical compounds that occurs naturally in plants and these compounds are medicinally active compounds which are dissolved out from the cells and tissue of plants by the process of extraction.

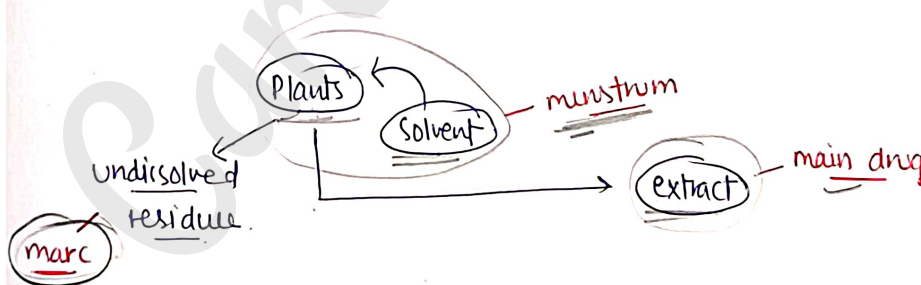


• Also it deals with the study of these phytoconstituents.

① Write a details note on Extraction.

• Extraction →

The process of extracting or separating medicinally active constituents of plants with the help of solvent and standard procedure is called as extraction.



- Solvent (menstruum) used for extraction

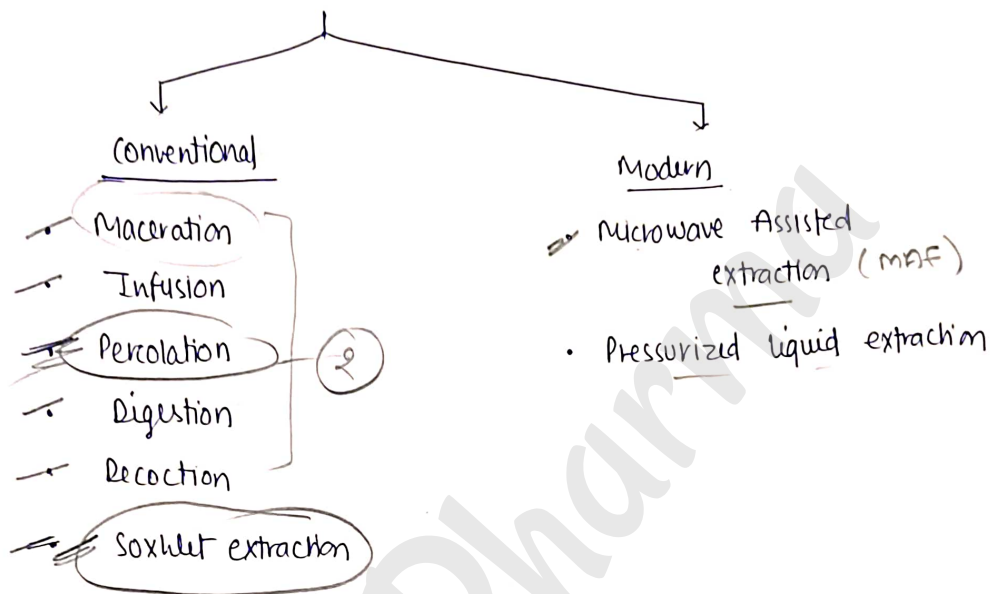
water, Ethanol, Methanol, chloroform, benzene, Dichloroethanol,
acetone, petroleum ether etc..

- It should be inert, non-toxic, inflammable etc.

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. Types of extraction →



i) Maceration → Solid plant materials are placed in closed soaking containers with solvent and allowed to stand for atleast 3 days with frequent agitation until soluble matter is dissolved, then mixture is strained and purified by filtration.



ii) Infusion → The plant material is macerated for a short period of time by using cold/boiling water

iii) Digestion → It is also a form of maceration by with gentle heat (40-60°C) is applied during the process of extraction
 • Also stir during extraction. eg, extract the morphine by applying heat.

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iv) Decoction →

Plant material is boiled in water then cooled and strained
this is suitable for water soluble and for heat stable
~~constituent~~ constituents.

v) Percolation →

It is a continuous downward displacement of solvent through bed, then crude drug material is carried out.

• procedure -

- Size reduction → firstly plant material is converted into coarse/fine powder through reducing their size.



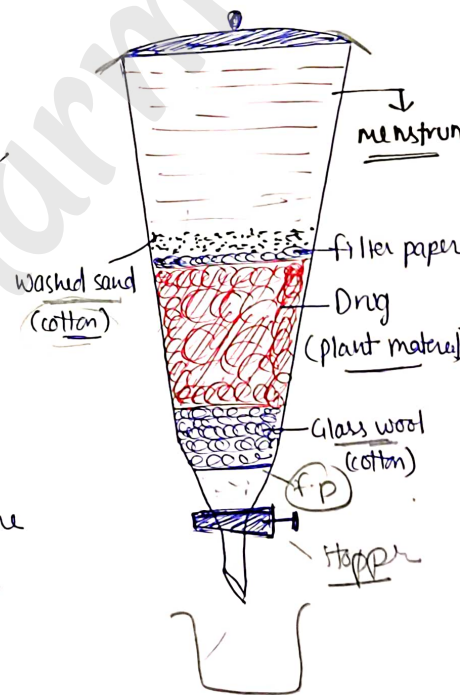
- Imbibition → process of absorption of one substance by another, in this powdered drug is moistened with suitable amount of menstrum (solvent) and allow to stand



- packing → After imbibition, the moisten drug is packed into percolator



- Maceration → sufficient amount of menstrum is added to saturated materials and allow to stands for 24 hrs macerate the drug.



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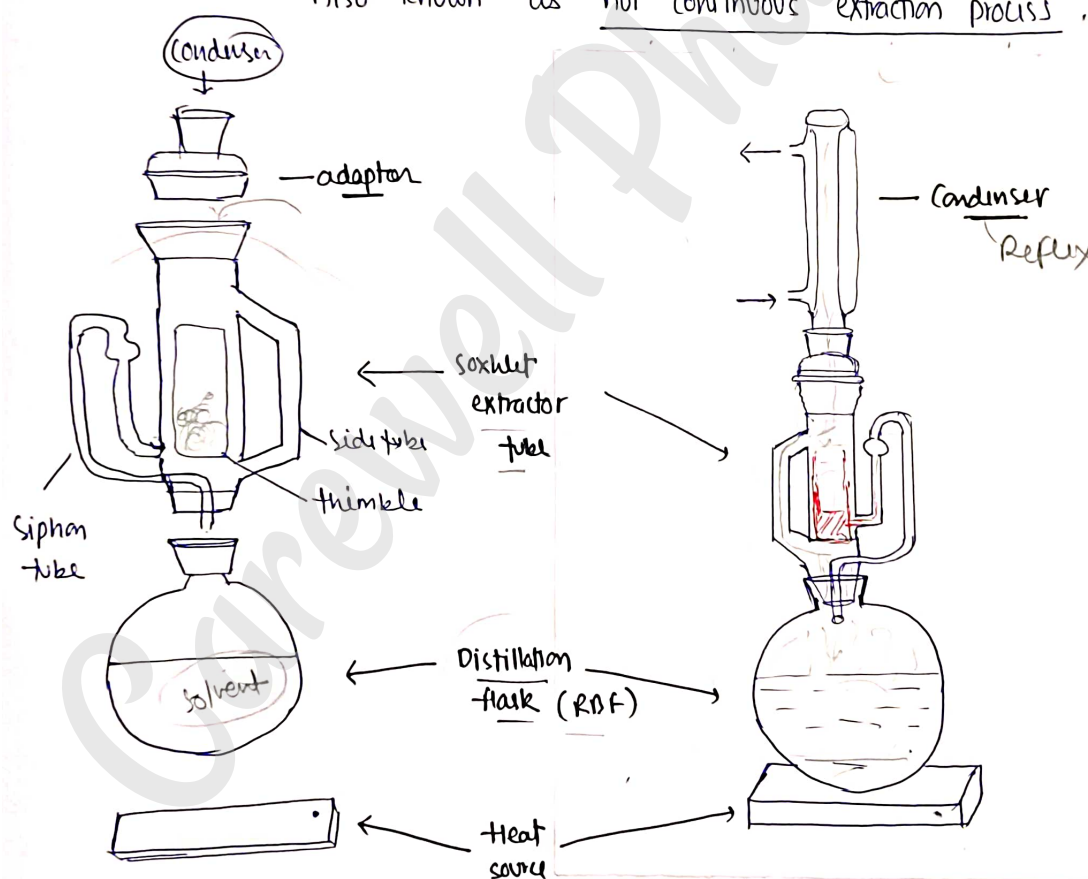
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- ↓
- Percolation → final step, where the ^{lower} top is opened and liquid is collected then it allow to drip slowly at controlled rate until $\frac{3}{4}$ volume of finished product is obtained.

• This process is repeated till a drop of solvent from the percolator when evaporator does not leave a residue.

(vi) Soxhlet extraction -

Also known as hot continuous extraction process.



- It is the best method for the continuous extraction of a solid by a hot solvent.

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• Construction & apparatus →

- RBC (Round bottom flask) or Distillation flask
- Soxhlet extraction tube (a thimble, side tube, siphon tube)
- Adaptor (to join condensor)
- Condensor (Reflex)

• procedure (working) →

firstly convert plant ~~material~~ into dry powdered form



Now, this powdered is placed in a thimble made up of filter paper, then it placed inside the ~~soxhlet~~ soxhlet apparatus.



This apparatus is fitted to a round bottom flask (RBF) which contain solvent (menstrum) and also with a reflex condenser with the help of adaptor.



The solvent in the RBF is boiled gently, vapours passes through the side tube, then it condensed through condenser and falls into thimble containing the material and slowly fills the soxhlet.



when the solvent reaches the top of the attached tube it siphons over into flask (move into RBF)

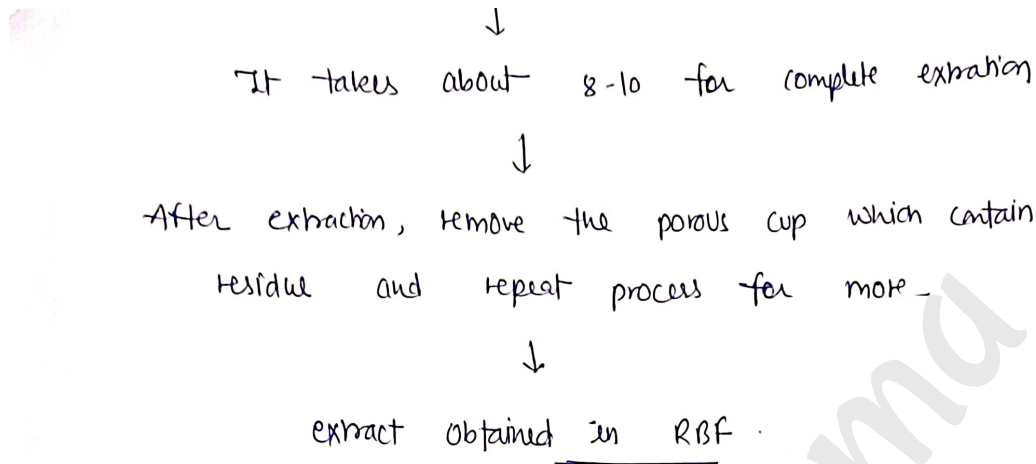


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Adv →

- required less time than percolation
- more economical
- performed on large scale

Also use other process for extraction such as distillation →

Steam distillation
Hydro distillation

• Modern methods

i) Microwave Assisted extraction (MAE) →

- extraction using microwaves -
- microwaves are non-ionizing electromagnetic waves.
(frequency 300 MHz - 300 GHz)

• procedure →

microwaves produced a high temperature

↓

evaporates the moisture in cells

↓

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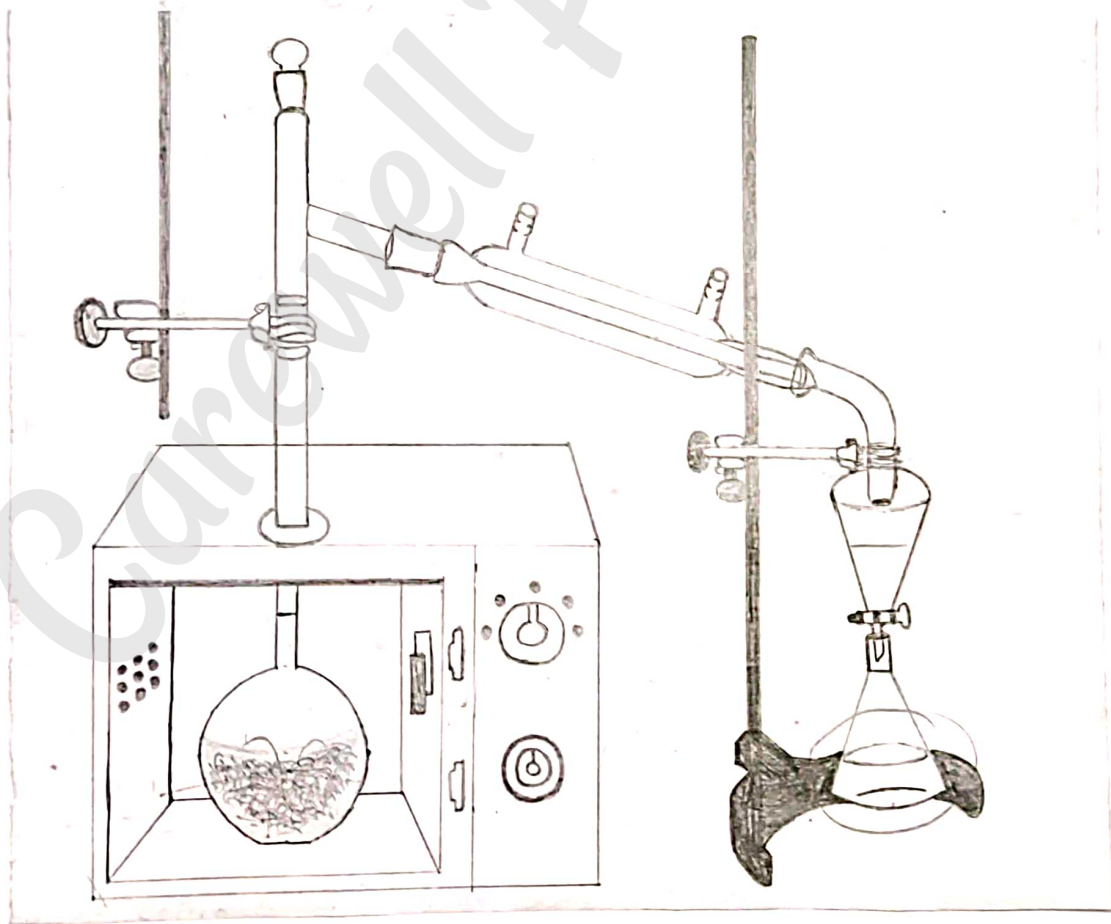
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↓
dehydrates cellulose
↓
ruptures cellwalls
↓
extract with solvent
↓
Collected with the help of ^{condenser.} ~~extraction.~~

- It has rapid extraction within 5-10 minutes.



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② Write about application of chromatography techniques in the isolation, purification and identification of crude drugs.

• chromatography →

It is a techniques which are used in the isolation, purification and identification of crude drugs.

OR

- It is a purification techniques, in which we have to identify the components and also separate it from mixtures.

Basic principle →

There are two main components (phases)

- stationary phase (It has large surface area and it may be solid / liquid)

- Mobile phase (It may be liquid / gas, It elute (flow) upon the stationary phase).

• Separation of the sample components is based on the difference in the rate of migration (movement) on stationary phase

⇒ Now, sample mixture is spotted on the stationary phase then eluted by mobile phase, mixture is separated on stationary phase

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Applications →

- Identification & Isolation
 - To identify the actual components in a complex mixtures.
 - To isolate (separate) the components from the mixture
- Analysis
 - To determine the quantity of active constituents present in mixture
 - Also helps to check the quality of ~~any~~ drugs by comparing with standard
- Purification
 - Overall, the main role of chromatography is to purify the drugs.

* There are several technique of chromatography ^{which are} used for this purpose -

i) paper chromatography

ii) TLC (Thin layer chromatography) or performance

iii) HPLC (High pressure liquid chromatography)

iv) HPTLC (High pressure thin layer chromatography)

v) Column chromatography.



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i) paper chromatography →

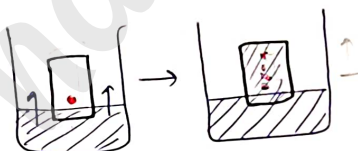
In this technique, sheet of paper is used to analysis of unknown substance.

- Cellulose layer in filter paper contain moisture which acts as stationary phase and Organic solvents/buffers & also (water) are used as mobile phase.

eg. (water) on (paper)

mobile phase

stationary phase



ii) TLC (Thin layer chromatography) →

It is used to identify the phytoconstituents and also used for qualitative determination of plant extracts.

- It is widely used technique and is similar to paper chromatography.
- It involves a stationary phase of paper a thin layer of adsorbent like silica gel, alumina or cellulose.

procedure →

Once a stationary phase plate is ready, draw a line using pencil 1cm from base of plate & also from top



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↓

Small spot of sample is ~~applied~~ applied on stationary phase then placed it in mobile phase

↓

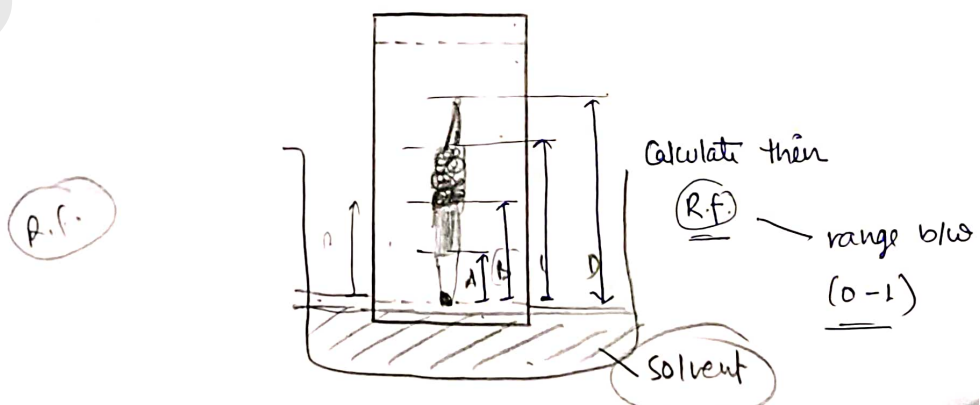
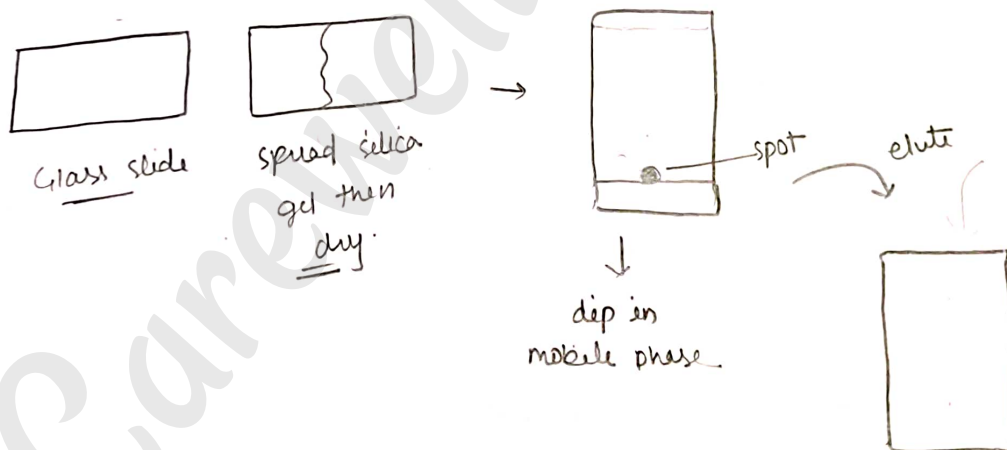
The mobile phase runs upward along the plate due to capillary action and the components get separated depending upon their polarity towards stationary phase

↓

Calculate retention factor

(Rf)

Distance travelled by an individual component ^(B)
 Distance travelled by the solvent. ^(S.D)



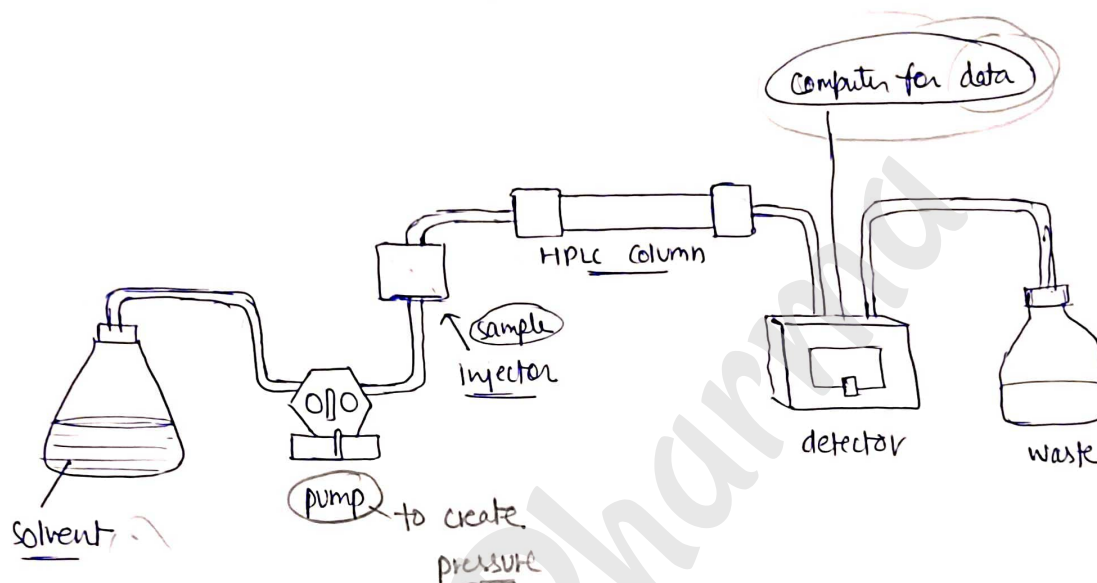
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iii) HPLC (High pressure/performance liquid chromatography) .



HPLC: high performance liquid chromatography

- Also called as high pressure liquid chromatography
- It relies on pumps to pass a pressurized (5000-1000 psi) liquid solvent i.e. mobile phase containing the sample mixture through a column filled with a solid adsorbent material.
- Stationary phase-immobilized thin layer liquid on micro glass or plastic beads tightly packed on narrow column
- Mobile phase- solvent system passed under high pressure through column
- The eluents can be detected by detectors
- It can be applied in the form of adsorption, ion exchange, partition or molecular sieve chromatography
- Due to rapidity in detection it is used for detection of amino acids, peptides, carbohydrates, proteins, lipids, nucleic acids, vitamins, hormones, drugs, etc

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iv) HPTLC (High pressure/performance thin layer chromatography)

- It has similar approach and employed the same physical principles of TLC (adsorption chromatography) i.e. the principle of separation is adsorption.

procedure →

sample is spotted on stationary phase then mobile phase solvent flow through because of capillary action.



components move according to their affinity towards the adsorbent.

- very less amount of sample is spotted on the plate



should not more than 1 mm in diameter

- It provides a higher efficacy than TLC because adsorbents used are small and uniform in size

(eg)

source - Tincture of Aloe vera var officinalis

stationary phase - silica gel

Mobile phase - Ethyl acetate : formic acid : water

(17:2:3)

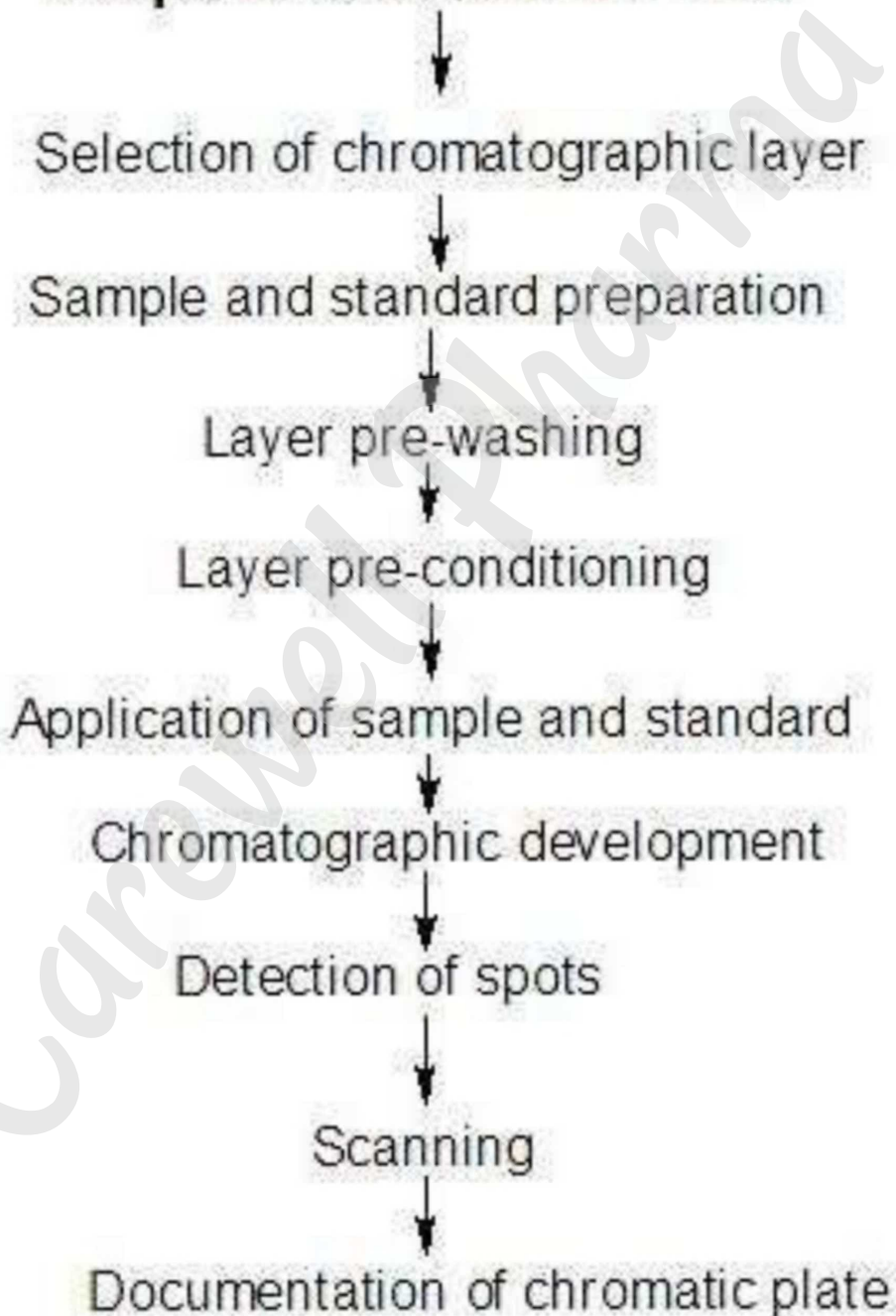
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Steps Involved in HPTLC



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